MILITARY SPECIFICATION

BODY, LOADED ASSEMBLY, GENERAL PURPOSE GRENADE, M73

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings for parts and assembly of the loaded body assembly of the M73 General Purpose H.E. Grenade.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1. Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-B-63469	Body Assemblies, General Purpose Grenade: M73 and M75 (Practice)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-453	Inspection, Radiographic
MIL-STD-1167	Ammunition Data Card

FSC 1330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

Code of Federal Regulations

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402; orders shall specify 49 CFR 100-199.)

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGE DRAWINGS

9334105	Body, Loaded Assembly
9345297	Box, Fiber Board, Packing, Ammunition for Grenade, GP: M73 Body Loaded Assembly
9345298	Packing and Marking for Wood Box for Grenade, GP: M73 Body Loaded Assembly

INSPECTION EQUIPMENT DRAWINGS

9362755	Detonator, Electric, PA526 (XM70)
9333825	Fuze, M230

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9334105 and associated drawings and with all requirements specified in applicable specifications and standards.

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3.3 Body assembly, M73. The body assembly shall comply with specification MIL-B-63469.

3.4 Radiographic quality. X-rays for evaluating the Composition B casting shall be taken in accordance with MIL-STD-453 to ascertain that there are no gross included voids or extraneous material. Radiographs shall be of a quality which will permit proper evaluation as determined by the responsible technical agency of the government.

3.5 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.5.4.

3.6 Reliability. The M73 Grenade shall demonstrate a minimum acceptable value (MAV) of 93.7 percent with a best operational capability (BOC) of 98 percent at 90 percent confidence.

3.7 First article inspection. This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. Unless otherwise specified on the drawing, all parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities for inspections. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the Contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are made a part of this specification.

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4.2 Classification of inspections. Unless otherwise specified herein, or in the contract, the following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Wave Shaper	9334108	15
Pellet, Booster	9334107	15
Body, Loading Assembly	9334103	15
Body, Loaded Assembly	9334105	229

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-64043(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Grenade, General Purpose: M73	1 OF 1	Listed Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Wave Shaper (Dwg. 9334108)				
	Examination for defects	15		3.2	4.4.2.1
	Pellet, Booster (Dwg. 9334107)				
	Examination for defects	15		3.2	4.4.2.2
	Body, Loading Assembly (Dwg. 9334103)				
	Examination for defects	15		3.2	4.4.2.3
	Radiographic examination	15		3.5	4.5.1
	Body, Loaded Assembly (Dwg. 9334105)				
	Examination for defects	25		3.2	4.4.2.4
	Functional Test	229		3.6	4.5.2

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Body, Loaded Assemblies shall contain:

a. Body assemblies from no more than two lots bearing the same interfix number from one manufacturer.

b. Booster pellets from lots bearing the same interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classfication of Defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-64043(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Wave Shaper	1 OF 1	9334108
			NEXT HIGHER ASSEMBLY 9334105

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
MINOR					
201	Flatness		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-64043(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Pellet, Booster	1 OF 1	9334107
			NEXT HIGHER ASSEMBLY 9334105

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
MINOR					
201	True position of center hole		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-64043(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Body, Loading Assy.	1 OF 1	9334103
			NEXT HIGHER ASSEMBLY: 9334105

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Presence of HE around fuze seat		100%	3.2	Visual
MAJOR					
101	Surface void limits of HE		0.40%	3.2	Visual/Gage
102	Diameter of waveshaper cavity		0.40%	3.2	Gage
103	Diameter of booster pellet cavity		0.40%	3.2	Gage
MINOR					
201	Depth of waveshaper cavity		0.65%	3.2	Gage
202	Depth of booster pellet cavity		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-64043(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Body, Loaded Assy.	1 OF 1	9334105
			NEXT HIGHER ASSEMBLY 9334143

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Functional test	125		3.6	4.5.2
MINOR					
201	Waveshaper missing		0.65%	3.2	Visual
202	Felt washer missing		0.65%	3.2	Visual
203	Booster pellet missing		0.65%	3.2	Visual
204	Felt pad missing or improperly installed		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.8	Visual

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4.4.3 Testing. Testing is described in the Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein. The contractor shall submit all inspection equipment designs to the government for evaluation and approval prior to fabrication of such equipment.

4.5 Test methods and procedures.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Radiographic. An X-Ray of each of the first article samples shall be made viewing axially. No included voids greater than 1/8 inch in diameter or more than three voids per quadrant shall exist. There shall be no evidence of foreign material in the casting.

4.5.2 Functional test. The Body, Loaded Assembly shall be functioned with PA526 detonator mounted in a plate which simulates the M230 Fuze Assembly. If two (2) or more assemblies do not exhibit high order detonation, the lot shall be rejected.

4.5.3 Data requirements. The contractor shall submit a quality inspection test demonstration and evaluation report in accordance with Data Item Descriptions DI-R-1721 and DI-R-1724 giving the detailed test results.

5. PACKAGING

5.1 Packing and marking. The Body, Loaded Assemblies shall be packed and marked in accordance with drawings 9345297 and 9345298, and shall comply with CFR49, Parts 100-199.

5.2 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 Intended use. The Body, Loaded Assemblies covered by this specification are intended for use in the M73 Grenade.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approval. See 6.2 of MIL-A-48078. Submit designs as required to Commander, Armament Research and Development Center, U.S. Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAF-I(D), Dover, NJ 07801.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
4.5.3	Quality Inspection Defect Report	DI-R-1721
4.5.3	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.5 First article. When a first article is required for inspection and approval, see 3.7 and 4.3.

6.6 Source control and specification control drawings. Certificates of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the government representative.

Custodian:
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Preparing activity:
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